



skill IT
EDUCATION

Data Science

PROGRAM BROCHURE —



Industry-Ready
Skills



Hands-on
Learning



Career
Support



Future-Focused
Growth



Analyze Today. Solve Tomorrow. Shape the Future.

Our mission is to empower learners with data-driven skills and real-world insights to solve complex problems and create meaningful impact across industries.

TABLE OF CONTENTS

<u>SkillIT Accolades</u>	>	2
<u>Why Skill IT</u>	>	3
<u>Real Time Internship</u>	>	4
<u>Job Ready Program</u>	>	5
<u>Program Curriculum</u>	>	7
<u>Contacts and Admission</u>	>	21

Skill IT Education

4+

Years of Excellence



10k+

Learners



योग: कर्मसु कौशलम्

IIT PATNA

Recognized for Excellence in
Professional Education & Training

NASSCOM®

NASSCOM

Aligned with Industry Standards
to Bridge the Skill Gap



SKILL IT INTERNSHIP CERTIFICATE

Awarded for Successful Completion
of Real-time Internship



INDUSTRY RECOGNITION

Trusted by Learners and
Industry Professionals



LEARNER TRUST

Built on Trust, Transparency
and Quality Education



CAREER FOCUSED

Empowering Learners to Build
Future-ready Careers



skill IT
EDUCATION

Data Science | Program Brochure | © Skill IT Education 2026

WHY Skill IT?

TOP 4 REASONS

1

Curriculum aligned with Industry

Syllabus aligned with industry as per global accreditation bodies



NASSCOM®

2

Ashok Veda as Mentor

Be mentored by industry-leading AI and Data Science experts committed to your career success



linkedin.com/in/ashokveda/

3

Realtime Internship

Gain real-world experience through internship opportunities in Analytics, Data Science, and Artificial Intelligence.

4

Top Placement records

Launch your career with confidence through our dedicated placement support and expert career guidance

KEY HIGHLIGHTS

1. Flexible Learning

Learners can repeat sessions, change batches, change learning modes, ad-hoc doubts sessions anytime.

2. Placement Assistance

A dedicated Career Assistance Team supports learners throughout their career journey, helping them achieve successful placements

3. Life-Time Access

Enjoy lifetime access to learning resources, enabling continuous learning and skill enhancement beyond course completion

4. Exclusive Practice Lab

Get exclusive access to an AI & Data Science Virtual Lab to practice concepts through hands-on learning

5. Job-oriented circular

Our curriculum is designed by industry experts to meet current market demands and prepare learners for successful careers

6. Elite Mentors

Learn from elite mentors and experienced faculty with real-world expertise from leading companies and premier institutions

7. Unlimited Projects

Unlimited projects with flexibility to choose from various industries but a minimum of 5 projects are required to complete projects phase.

8. Learning Community

Join an exclusive online learning community with active learners, mentors, and alumni for continuous guidance and support.

REAL-TIME INTERNSHIP

REAL-WORLD EXPERIENCE THROUGH INTERNSHIPS & INDUSTRY PROJECTS

At Skill IT Education, learners gain practical experience by working on real-world industry projects and hands-on assignments aligned with current market requirements.

Our project-based learning approach enables learners to apply classroom concepts to real business scenarios under the guidance of experienced mentors and industry professionals. This practical exposure helps build technical expertise, problem-solving abilities, and the confidence required for successful careers in the IT industry.

internship@skillit.com



JOB READY PROGRAM

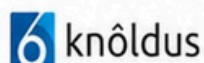
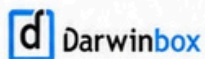
COMPLETE CAREER TRANSITION SUPPORT

Skill IT Education's Placement Assistance Team works closely with every learner, offering personalized career guidance, interview preparation, and placement support to ensure a smooth transition into the IT industry



Our Top Hiring Partners

Skill IT Education has a strong network of **1000+** hiring partners





PROGRAM CURRICULUM

DATA SCIENCE – COURSE BUNDLE

- Skill IT Education Data Science – Course Bundle is one of the **world's most popular, comprehensive, job-oriented, advanced** Data Science courses designed to help learners build future-ready careers.
- The course is **rigorously updated** as per industry requirements and fine-tuned to make the learning process structured and effective for real-world applications.



LEARNING HOURS

280 Hrs



TOTAL DURATION

7 Months



ADD-ON

Internship, Placements

LEARNING HOURS	COURSE	HOURS
1	Data Science Foundation	14 Hrs
2	Python Foundation	28 Hrs
3	Statistics Essentials	14 Hrs
4	Machine Learning Associate	28 Hrs
5	Machine Learning Expert	28 Hrs
6	Advanced Data Science	28 Hrs
7	Database: SQL and MongoDB	28 Hrs
8	Version Control with Git	14 Hrs
9	Big Data Foundation	14 Hrs
10	Certified Business Intelligence (BI) Analyst	14 Hrs



skill IT
EDUCATION

DATA SCIENCE FOUNDATION



PREREQUISITES

Python Foundation



LEARNING HOURS

16 Hrs

MODULE 1

DATA SCIENCE ESSENTIALS

- Introduction to Data Science
- Evolution of Data Science
- Data Science vs AI/Machine Learning
- Data Science vs Analytics
- Big Data Vs Data Science
- Data Science Terminologies

MODULE 2

DATA SCIENCE DEMO

- Data Preparation
- Business Requirement: Use Case
- Prediction with ML model
- Machine learning Model building
- Delivering Business Value

MODULE 3

ANALYTICS CLASSIFICATION

- Types of Analytics
- Descriptive Analytics
- Diagnostic Analytics
- Predictive Analytics
- Prescriptive Analytics
- EDA and insight gathering demo in Tableau

MODULE 4

DATA SCIENCE AND RELATED FIELDS

- Introduction to AI
- Introduction to Computer Vision
- Introduction to Natural Language Processing
- Introduction to Reinforcement Learning
- Introduction to GAN
- Introduction to Generative Passive Models

MODULE 5

DATA SCIENCE ROLES & WORKFLOW

- Data Science Project workflow
- Data Science Project stages.
- Roles: Data Engineer, Data Scientist, ML Engineer and MLOps Engineer

MODULE 6

MACHINE LEARNING INTRODUCTION

- What Is ML? ML Vs AI
- Supervised Vs Unsupervised
- ML Workflow, Popular ML Algorithms
- Clustering, Classification And Regression

MODULE 7

DATA SCIENCE INDUSTRY APPLICATIONS

- Data Science in Finance and Banking
- Data Science in Retail
- Data Science in Health Care
- Data Science in Logistics and Supply Chain
- Data Science in Technology Industry
- Data Science in Manufacturing
- Data Science in Agriculture

TOOLS/PLATFORMS COVERED



skill IT
EDUCATION



PYTHON FOUNDATION



PREREQUISITES

None



LEARNING HOURS

32 Hrs

MODULE 1

PYTHON BASICS

- Introduction of python
- Installation of Python and IDE
- Python Variables
- Number & Booleans, strings
- Python basic data types
- Arithmetic Operators
- Assignment Operators
- Comparison Operators

MODULE 2

PYTHON CONTROL STATEMENTS

- IF Conditional statement
- NESTED IF
- IF-ELSE
- Python Loops basics
- FOR statements
- WHILE Statement
- BREAK and CONTINUE statements

MODULE 3

PYTHON DATA STRUCTURES

- Basic data structure in python
- Tuple: Object, methods
- List: Object, methods
- Sets: Object, methods
- Dictionary: Object, methods

MODULE 4

PYTHON FUNCTIONS

- Functions basics
- Lambda functions
- Function Parameter passing
- Map, reduce, filter functions

TOOLS/PLATFORMS COVERED



skill IT
EDUCATION

STATISTICS ESSENTIALS



PREREQUISITES

None



LEARNING HOURS

20 hrs.

MODULE 1

OVERVIEW OF STATISTICS

- Introduction to Statistics
- Basic Terms Of Statistics
- Types Of Data
- Descriptive And Inferential Statistics

MODULE 2

HARNESSING DATA

- Types of Sampling
- Random Sampling
- Simple Random Sampling
- Systematic Random Sampling
- Stratified Random Sampling
- Cluster Random Sampling
- Multi stage Sampling
- Sampling With Replacement And Without Replacement
- Sampling Error
- Cochran's Minimum Sample Size
- Methods Of Collecting Data

MODULE 3

EXPLORATORY DATA ANALYSIS

- Exploratory Data Analysis Introduction
- Measures Of Central Tendencies: Mean, Median And Mode
- Measures Of Central Tendencies: Range, Variance And Standard Deviation
- Data Distribution Plot: Histogram
- Normal Distribution & Properties
- Central Limit Theorem
- Skewness & Kurtosis
- Z Value / Standard Value
- Empirical Rule and Outliers
- Normality Testing
- Measures Of Distance: Euclidean, Manhattan And Minkowski Distance
- Covariance & Correlation

MODULE 4

HYPOTHESIS TESTING

- Hypothesis Testing Introduction
- Types of Hypothesis Testing
- P- Value, Critical Region
- Hypothesis Testing Errors : Type I And Type II
- Two Sample Independent T-test
- Two Sample Relation T-test
- One Way Anova Test
- Application of Hypothesis testing



MACHINE LEARNING ASSOCIATE



PREREQUISITES

Python Foundation, DSF



LEARNING HOURS

40 Hrs

MODULE 1

MACHINE LEARNING INTRODUCTION

- What Is ML? ML Vs AI
- Clustering, Classification And Regression
- Supervised Vs Unsupervised

MODULE 2

PYTHON NUMPY PACKAGE

- Introduction to Numpy Package
- Core Numpy functions
- Array as Data Structure
- Matrix Operations, Broadcasting in Arrays

MODULE 3

PYTHON PANDAS PACKAGE

- Introduction to Pandas package
- File Reading in Pandas
- Data Frame in Pandas
- Series in Pandas
- Data munging with Pandas

MODULE 4

VISUALIZATION WITH PYTHON - Matplotlib

- Visualization Packages (Matplotlib)
- Basic Plots: Line, Bar, Pie, Scatter
- Components Of A Plot, Sub-Plots

MODULE 5

PYTHON VISUALIZATION PACKAGE - SEABORN

- Advanced Python Data Visualizations
- Seaborn: Basic Plot

MODULE 6

ML ALGO: LINEAR REGRESSION

- Introduction to Linear Regression
- Modeling and Evaluation in Python
- How it works: Regression and Best Fit Line

MODULE 7

ML ALGO: LOGISTIC REGRESSION

- Introduction to Logistic Regression
- Modeling and Evaluation in Python
- How it works: Classification & Sigmoid Curve

MODULE 8

ML ALGO: K MEANS CLUSTERING

- K Means Algorithm
- Modeling in Python
- How it works : K Means theory
- Understanding Clustering (Unsupervised)

MODULE 9

ML ALGO: KNN

- How It Works: Nearest Neighbor Concept
- Modeling and Evaluation in Python
- Introduction to KNN



python



pandas



NumPy

matplotlib



edabit learn



Google colab



skill IT
EDUCATION



MACHINE LEARNING EXPERT



PREREQUISITES

Python Foundation, DSF



LEARNING HOURS

40 Hrs

MODULE 1

FEATURE ENGINEERING

- Introduction to Feature Engineering
- Use case for feature engineering
- Feature Engineering Techniques: Encoding, Scaling, Data Transformation
- Handling Missing values, handling outliers
- Creation of Pipeline

MODULE 2

ML ALGO: SUPPORT VECTOR MACHINE (SVM)

- Introduction to SVM
- Modeling and Evaluation of SVM in Python
- How It Works: SVM Concept, Kernel Trick

MODULE 3

PRINCIPAL COMPONENT ANALYSIS (PCA)

- Building Blocks Of PCA
- Modeling PCA in Python
- How it works: Finding Principal Components

MODULE 4

ML ALGO: DECISION TREE

- Introduction to Decision Tree & Random Forest
- Modeling and Evaluation in Python
- How it works

MODULE 5

ENSEMBLE TECHNIQUES - BAGGING

- Introduction to Ensemble technique
- Modeling and Evaluation in Python
- Bagging and How it works

MODULE 6

ML ALGO: NAÏVE BAYES

- Introduction to Naive Bayes
- Modeling and Evaluation in Python
- How it works: Bayes' Theorem
- Naive Bayes For Text Classification

MODULE 7

GRADIENT BOOSTING, XGBOOST

- Introduction to Boosting and XGBoost
- Modeling and Evaluation of in Python
- How it works?

TOOLS/PLATFORMS COVERED





PREREQUISITES

Python Foundation, MLE, DSF



LEARNING HOURS

40 Hrs

MODULE 1

TIME SERIES FORECASTING - ARIMA

- What is Time Series?
- Autoregressive Model (AR)
- Moving Average Model (MA)
- Time Series Analysis in Python
- Trend, Seasonality, cyclical and random
- Stationarity of Time Series
- ARIMA Model
- Autocorrelation and AIC

MODULE 2

SENTIMENT ANALYSIS

- Introduction to Sentiment Analysis
- Case study: Sentiment Analysis on Movie Reviews
- NLTK Package

MODULE 3

REGULAR EXPRESSIONS WITH PYTHON

- Regex Introduction
- Text extraction with Python Regex
- Regex codes

MODULE 4

ML MODEL DEPLOYMENT WITH FLASK

- Introduction to Flask
- Flask application – ML Model deployment
- URL and App routing

MODULE 5

ADVANCED DATA ANALYSIS WITH MS EXCEL

- MS Excel core Functions
- Data Table
- Pivot Table
- Advanced Functions (VLOOKUP, INDIRECT..)
- Linear Regression with EXCEL
- Goal Seek Analysis
- Solving Data Equation with EXCEL

MODULE 6

AWS CLOUD FOR DATA SCIENCE

- Introduction of cloud
- AWS Service (EC2 instance)
- Difference between GCC, Azure,AWS

MODULE 7

AZURE FOR DATA SCIENCE

- Introduction to AZURE ML studio
- Data Pipeline
- ML modeling with Azure

MODULE 8

INTRODUCTION TO DEEP LEARNING

- Introduction to Artificial Neural Network, Architecture
- Introduction to Convolutional Neural Network, Architecture
- Artificial Neural Network in Python
- Convolutional Neural Network in Python

TOOLS/PLATFORMS COVERED



Amazon
SageMaker



Azure
Machine
Learning



Flask
web development,
one drop at a time

Natural Language
ToolKit

DATABASE: SQL AND MONGODB



PREREQUISITES

None



LEARNING HOURS

40 Hrs

MODULE 1

DATABASE INTRODUCTION

- DATABASE Overview
- Key concepts of database management
- Relational Database Management System
- CRUD operations

MODULE 2

SQL BASICS

- Introduction to Databases
- Introduction to SQL
- SQL Commands
- MY SQL workbench installation

MODULE 3

DATA TYPES AND CONSTRAINTS

- Numeric, Character, date time data type
- Primary key, Foreign key, Not null
- Unique, Check, default, Auto increment

MODULE 4

DATABASES AND TABLES (MySQL)

- Create database
- Show and use databases
- Create new table from existing data types
- Delete table, Delete table records
- Create table, Rename table
- Insert into, Update records
- Alter table

MODULE 5

SQL JOINS

- Create database
- Delete database
- Show and use databases
- Create table, Rename table
- Delete table, Delete table records
- Create new table from existing data types
- Insert into, Update records
- Alter table

MODULE 6

SQL COMMANDS AND CLAUSES

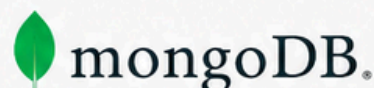
- Select, Select distinct
- Aliases, Where clause
- Relational operators, Logical
- Between, Order by, In
- Like, Limit, null/not null, group by
- Having, Sub queries

MODULE 7

DOCUMENT DB/NO-SQL DB

- Introduction of Document DB
- Document DB vs SQL DB
- Popular Document DBs
- MongoDB basics
- Data format and Key methods

TOOLS/PLATFORMS COVERED





VERSION CONTROL WITH GIT



PREREQUISITES

None



LEARNING HOURS

20 Hrs

MODULE 1

GIT INTRODUCTION

- Purpose of Version Control
- Popular Version control tools
- Git Distribution Version Control
- Terminologies
- Git Workflow
- Git Architecture

MODULE 2

GIT REPOSITORY and GitHub

- Git Repo Introduction
- Create New Repo with Init command
- Git Essentials: Copy & User Setup
- Mastering Git and GitHub

MODULE 3

COMMITTS, PULL, FETCH AND PUSH

- Git Repo Introduction
- Create New Repo with Init command
- Git Essentials: Copy & User Setup
- Mastering Git and GitHub

MODULE 4

TAGGING, BRANCHING AND MERGING

- Organize code with branches
- Checkout branch
- Merge branches

MODULE 5

UNDOING CHANGES

- Editing Commits
- Commit command Amend flag
- Git reset and revert

MODULE 6

GIT WITH GITHUB AND BITBUCKET

- Creating GitHub Account
- Local and Remote Repo
- Collaborating with other developers
- Bitbucket Git account

TOOLS/PLATFORMS COVERED





BIG DATA FOUNDATION



PREREQUISITES

Python Foundation



LEARNING HOURS

20 Hrs

MODULE 1

BIG DATA INTRODUCTION

- What is Big Data and Hadoop
- Introduction to Hadoop
- Big Data Overview
- Five Vs of Big Data
- Components of Hadoop Ecosystem
- Big Data Analytics Introduction

MODULE 2

HDFS AND MAP REDUCE

- Key Terms: Output Format, Partitioners, Combiners, Shuffle, and Sort
- Distributed Processing with Map Reduce
- HDFS – Big Data Storage
- Mapping and reducing stages concepts

MODULE 3

PYSPARK FOUNDATION

- Spark Configuration
- PySpark Introduction
- Working with RDDs in PySpark
- Resilient distributed datasets (RDD)
- Aggregating Data with Pair RDDs

MODULE 4

SPARK SQL and HADOOP HIVE

- Spark SQL vs Hadoop Hive
- Introducing Spark SQL

TOOLS/PLATFORMS COVERED



skill IT
EDUCATION



CERTIFIED BI ANALYST



PREREQUISITES

None



LEARNING HOURS

20 Hrs

MODULE 1

TABLEAU FUNDAMENTALS

- Introduction to Business Intelligence & Introduction to Tableau
- Create Table Calculations
- Create Calculated Fields
- Create Custom Hierarchies
- Interface Tour, Data visualization: Pie chart, Column chart, Bar chart.
- Bar chart, Tree Map, Line Chart
- Area chart, Combination Charts, Map
- Dashboards creation, Quick Filters

MODULE 2

POWER-BI BASICS

- Power BI Introduction
- Basics Visualizations
- Dashboard Creation
- Basic Data Cleaning
- Basic DAX FUNCTION

MODULE 3

DATA TRANSFORMATION TECHNIQUES

- Exploring Query Editor
- Data Cleansing and Manipulation:
- Creating Our Initial Project File
- Connecting to Our Data Source
- Editing Rows
- Changing Data Types
- Replacing Values

MODULE 4

CONNECTING TO VARIOUS DATA SOURCES

- Connecting to a CSV File
- Connecting to a Webpage
- Extracting Characters
- Splitting and Merging Columns
- Creating Columns from Examples
- Creating Conditional Columns
- Create Data Model

TOOLS/PLATFORMS COVERED



skill IT
EDUCATION



CONTACTS & ADMISSION

Data Science – PROGRAM ENQUIRY



DURATION

: 7 Months



LEARNING MODE

: Live Online / In-Person Classroom (Selected Cities)



24x7 live chat www.skillit.com | admission@skillit.com



INDIA :



The best way to predict the future
is to **create it**.

– Peter Drucker



Data is the new oil.
Knowledge is the **new power**.

– Clive Humby



The goal is to turn data into information,
and information into **insight**.

– Carly Fiorina



THE FUTURE IS SHAPED BY THOSE
WHO KEEP LEARNING TODAY.



LEARN DATA SCIENCE. BUILD SKILLS. CREATE IMPACT.

TAKE YOUR FIRST STEP TOWARDS A DATA SCIENCE CAREER

skill IT
EDUCATION